Entity-Token World Models and Differentiable Predictive Control for Building-Agnostic HVAC Control

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Abstract

A single controller for the Heating, Ventilation and Air Conditioning (HVAC) systems of unseen buildings has to cope with a different number of zones, air handlers and thermal properties in every building and with climates ranging from subarctic winters to tropical summers. This work addresses the Smart Buildings Challenge with a policy that learns its own control logic, without deep reinforcement learning and without imitating an existing controller. Every zone, air handler, hot-water system and the building as a whole is represented as a token, and a transformer whose attention is biased only by the structural relation between two tokens is used both as a learned world model of the building and as the control policy, thereby making both networks independent of the building size. The policy is trained by Differentiable Predictive Control (DPC), that is, by backpropagating the competition reward, rebuilt from the predicted temperatures and power, through closed-loop rollouts of up to 24 hours inside the world model, while the world model is refined in successive on-policy rounds. On the 30 official scenarios evaluated over 14 days, the final policy reduces the mean absolute step reward from 0.0591 for the shipped PID baseline to 0.0106, and it reaches 0.013 on the Kaggle public leaderboard. The experiments further indicate that the calibration of the world model on the reward terms matters more than its average temperature error.

1 Introduction

Buildings account for a large share of energy use and carbon emissions, and their HVAC systems are still mostly operated by rule-based logic tuned by hand for each building [4, 13]. Model predictive control and reinforcement learning have repeatedly shown large savings on individual buildings, but both require a model or a training process specific to the building at hand, and this cost has so far prevented their deployment at scale [11]. The Smart Buildings Challenge [7], built on the Smart Buildings Control Suite simulator [6], makes this obstacle explicit by asking for one policy that controls buildings never seen during development, with different floor plans, numbers of zones and air handlers, envelope materials and weather, and that is scored on a reward combining occupant comfort, energy cost and carbon emissions.

The approach presented here rests on two choices. First, the building is described as a set of entities (zones, air handlers, the hot-water system and a global token for weather and time), and the same transformer architecture, with attention biased by the structural relation between entities, is used for a world model of the building and for the policy, so that both apply to any building without modification.

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Second, the policy is not trained by trial and error on the simulator, nor by imitation of a baseline controller, but by DPC [5]: the competition reward is rebuilt from the world-model predictions and maximized by backpropagation through closed-loop rollouts, thereby letting the policy derive its own control logic from the physics captured by the world model. The main contributions of this work are the following:

- an entity-token world model with long thermal memory, refined in on-policy rounds, with a closed-loop temperature error of 0.82 K after 24 hours on unseen buildings;
- a DPC scheme with a 24-hour horizon and a command filter inside the rollouts, which reaches 0.0106 locally and 0.013 on Kaggle, against 0.0591 for the shipped baseline;
- an analysis showing that the calibration of the world model on the reward terms, rather than its average error, determines the quality of the policy.

2 Related Works

The Smart Buildings Challenge [7] is the first benchmark in which a single HVAC policy is evaluated on unseen buildings generated at scale, using the simulator of the Smart Buildings Control Suite [6]; earlier benchmarks such as BOPTEST [2] provide detailed but individual test cases. Model Predictive Control (MPC) is the reference for advanced building control [4], although its dependence on a model identified for every building limits its scalability. Deep reinforcement learning has been applied to HVAC control since [14], with sample efficiency, safety and transfer to new buildings as its main obstacles [13, 11].

Differentiable control combines the two families. Differentiable MPC [1] embeds an optimization problem in a policy, and Gnu-RL [3] applies it to building HVAC control with a linear model; DPC [5] instead trains an explicit neural policy offline by differentiating a control objective through a model of the system, and it is the formulation adopted in this work. Learned world models [8, 9] allow policies to be trained in imagination, and the exploitation of model errors by the policy, discussed in [10], is the main failure mode encountered in the present experiments as well; short-horizon actor-critic methods such as SHAC [15] address long horizons with a learned value function, whereas the present work relies on an accurate world model and on gradient damping to differentiate directly through horizons of 24 hours. The entity-token architecture follows the transformer [12], and the attention bias depending on the relation between two nodes is inspired by Graphormer [16].

3 Methods

3.1 Problem Setting

At every 5-minute step t, the policy receives the observation o_t of a building with V zones and K air handlers, which contains the temperatures, setpoints and actuator readbacks of every zone, the supply-air, airflow, pressure and outdoor-air measurements of every air handler, the supply-water and pressure measurements of the hot-water system, the comfort-mode flags, the number of occupants and the time, and it returns the action a_t, made of the damper and reheat-valve commands $(u_v^{\mathrm{d}}, u_v^{\mathrm{r}})$ of every zone v, the run state, supply-air temperature, static pressure and recirculation commands of every air handler k, and the run state, supply-water temperature and differential pressure of the hot-water system. The step reward is

$$r_t = w_p\, R_t - w_e\, \tilde{E}_t - w_c\, \tilde{C}_t, \tag{1}$$

where $R_t \leq 0$ is the productivity regret of the occupants, which grows linearly and then quadratically with the deviation $\delta_{v,t} = \max(T_{v,t} - T_{v,t}^{\mathrm{c}},\, T_{v,t}^{\mathrm{h}} - T_{v,t},\, 0)$ of every occupied zone from its comfort band, and $\tilde{E}_t$ and $\tilde{C}_t$ are the normalized energy cost and carbon emissions of electricity and natural gas. The score of a policy is the mean absolute step reward $S = \frac{1}{N}\sum_t |r_t|$ over all steps and scenarios, so that lower is better.

3.2 Entity-Token Architecture

The architecture shared by the world model and the policy is outlined in Figure 1.

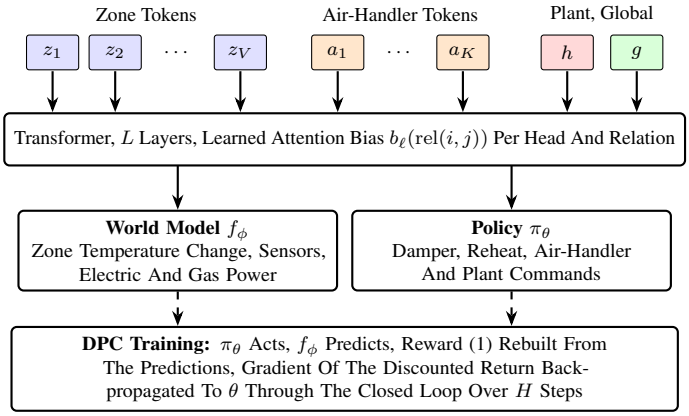


Figure 1: Entity-token architecture shared by the world model and the policy (same structure, separate weights), and the DPC training loop that connects them.

As can be observed in Figure 1, a building is represented by $N = V + K + 2$ tokens, one per zone, one per air handler, one for the hot-water system and one global token. The input vector of zone v concatenates its normalized observations, its last three temperature changes, its previous commands and seven static descriptors g_v extracted once from the floor plan, namely the logarithm of the floor area, the exterior-wall fraction, the exterior wall per square root of area, the floor position, the ground-floor and top-floor flags and the number of zones served by the same air handler; air-handler and hot-water tokens concatenate their sensors, including the outdoor air temperature measured by every air handler, with their previous commands, while the global token carries the comfort-mode flags, the number of occupants, the hour and day of the week and, for the policy, the hours to the next scheduled comfort period, the hours left in the current one and a working-day flag. Each token type $c \in \{\mathrm{zone}, \mathrm{ahu}, \mathrm{hws}, \mathrm{glob}\}$ has its own two-layer encoder and a learned type embedding,

$$x_i^{(0)} = \mathrm{MLP}_c(\mathrm{in}_i) + e_c, \tag{2}$$

after which L pre-norm transformer layers [12] process all tokens jointly. The only structural information enters through the attention logits: each pair of tokens (i, j) belongs to one of ten relations $\rho(i, j)$ (two zones on the same air handler or on different ones, a zone and its own or another air handler in both directions, two air handlers, any token and the hot-water system, any token and the global token, and a token with itself), and every head h of layer ℓ adds a learned scalar to the corresponding logit,

$$A_{ij}^{(\ell,h)} = \mathrm{softmax}_j\left(\frac{q_i^{(\ell,h)} \cdot k_j^{(\ell,h)}}{\sqrt{d_h}} + b_{\rho(i,j)}^{(\ell,h)}\right), \tag{3}$$

in the spirit of Graphormer [16], with the biases initialized to zero. Since no parameter depends on V or K, the same weights apply to every building. Separate linear heads read the output tokens: continuous commands pass through a hyperbolic tangent, while on/off commands use a hard threshold in the forward pass and the gradient of the underlying sigmoid in the backward pass (straight-through estimator).

3.3 World Model

Given the state s_t (the token inputs above), the action a_t and the exogenous inputs x_t read from the episode (weather, setpoints, occupancy, tariffs and carbon intensities), the world model f_ϕ predicts the normalized temperature change of every zone, the next actuator readbacks of every zone, the next sensor values of every air handler and of the hot-water system, and the electric and gas power of the building, the latter as sums of non-negative contributions of the individual tokens,

$$\begin{aligned} &\hat{T}_{v,t+1} = \hat{T}_{v,t} + \sigma_{\Delta T}\,\Delta\hat{T}_{v,t}, \\ &\hat{P}_t^{\mathrm{el}} = \textstyle\sum_k \mathrm{softplus}(y_k^{\mathrm{el}}) + \mathrm{softplus}(y_{\mathrm{hws}}^{\mathrm{el}}), \qquad \hat{P}_t^{\mathrm{gas}} = \textstyle\sum_v \mathrm{softplus}(y_v^{\mathrm{gas}}) + \mathrm{softplus}(y_{\mathrm{hws}}^{\mathrm{gas}}), \end{aligned} \tag{4}$$

from which the reward (1) is rebuilt exactly. Since walls, slabs and interior mass store heat that is not observed, the air temperature and its last changes cannot distinguish a zone with cold walls after a winter weekend from one with warm walls at the same air temperature; each zone token is therefore augmented by a long memory of exponential averages,

$$m^{(\tau)}_{v,t+1} = e^{-1/\tau}\, m^{(\tau)}_{v,t} + \big(1 - e^{-1/\tau}\big)\, \hat{T}_{v,t+1}, \tag{5}$$

of its temperature over $\tau = 12$, 72 and 288 steps (1, 6 and 24 hours), of its reheat and damper commands over 6 and 24 hours and of the outdoor air over 6 and 24 hours, which the rollout updates from its own predictions.

The model is trained on closed-loop unrolls of $U \in \{48, 144, 288\}$ steps (4, 12 and 24 hours) in which its own predictions are fed back while the logged actions are replayed, by minimizing

$$\mathcal{L}(\phi) = \frac{1}{U}\sum_{k=1}^{U}\Big(\frac{\sum_v \omega_v(\hat{T}_{v,k} - T_{v,k})^2}{\sigma^2_{\Delta T}\sum_v \omega_v} + w_R\big(\hat{R}_k - R_k\big)^2/\sigma^2_R + w_s\,\mathcal{L}^{\text{sens}}_k + w_P\,\big\|\hat{P}_k - P_k\big\|^2\Big), \tag{6}$$

where the zone weight $\omega_v = 1 + 3 \cdot \mathbb{1}[\text{occupied and within 1 K of or outside the band}]$ concentrates the accuracy where the reward is sensitive to temperature errors, the second term matches the comfort part of the reward directly, so that the error is measured in the units used by the DPC objective, with $\sigma_R = 0.1$, and $\mathcal{L}^{\text{sens}}$ collects the errors on readbacks and sensors; unrolls longer than 4 hours are recomputed in the backward pass (gradient checkpointing), so that memory does not grow with U.

The training data come from about 140 buildings produced by the official generator, with seeds disjoint from all evaluation sets, under six weather stations (Atlanta, Denver, Fairbanks, Minneapolis, Phoenix and Singapore). The first episodes were collected with an exploration policy switching every few hours between the shipped PID controller, its noisy version, randomly held setpoints and extreme plant states; afterwards, training alternates between the policy and the world model in on-policy rounds, summarized in Algorithm 1: in every round the current policy runs one long episode per building, mostly as is and partly with action noise or short idle and randomly held segments, and the world model is fine-tuned with half of each batch drawn from the new episodes, thereby correcting it precisely in the states that the policy visits. Five successive world models are compared in Section 4: WM-1 and WM-2 come from the exploration data and the first rounds, WM-3 introduces the memory (5) and the 24-hour unrolls, WM-4 adds a round of the first filtered policy, and WM-5 a round in which a quarter of the segments hold the plant setpoints at random levels.

3.4 Differentiable Predictive Control

The policy is trained by DPC [5], as detailed in the inner loop of Algorithm 1. From a batch of true start states s_{t_0} drawn from the collected episodes, the policy acts on the predicted states, the world model predicts the next state with the exogenous inputs of the episode and the reward is rebuilt from the prediction, so that the objective

$$J(\theta) = \mathbb{E}_{s_{t_0}}\Big[\frac{1}{H}\sum_{k=0}^{H-1}\gamma^k\,\hat{r}_{t_0+k}\Big], \qquad \hat{r}_t = \kappa\, w_p \hat{R}_t - w_e\hat{\bar{E}}_t - w_c\hat{\bar{C}}_t, \tag{7}$$

is maximized by backpropagation through the whole closed loop, where the factor $\kappa > 1$ on the comfort term counteracts the tendency of the policy to trade comfort for energy inside the world model. Three mechanisms make the optimization over horizons of hundreds of steps stable and transferable to the simulator. First, the gradient carried by the predicted state from one step to the previous one is damped without changing its value,

$$\hat{s}_{k+1} \leftarrow \mathrm{sg}(\hat{s}_{k+1}) + \lambda\big(\hat{s}_{k+1} - \mathrm{sg}(\hat{s}_{k+1})\big), \tag{8}$$

where sg stops the gradient, so that the credit assigned to an action by a reward h steps later is weighted by λ^h and the backpropagation does not explode. Second, since the policy otherwise reacts to world-model errors with fast oscillating commands, which the real plant does not reward, every continuous command is filtered both inside the rollouts and at deployment,

$$\tilde{u}_t = \alpha\, u_t + (1 - \alpha)\,\tilde{u}_{t-1}, \tag{9}$$

and the policy receives the filtered commands as its previous actions, exactly as in the building. Third, the horizon follows a curriculum from 12 to 24 hours, which is long enough for the policy to

experience the effect of preheating before occupancy and of letting the building drift at night and over the weekend. The start states exclude the generated buildings larger than the official size range, and every policy is warm-started from the policy of the previous round. The architecture and training settings of both networks are listed in Table 1.

Table 1: Architecture and training settings of the world model and of the final policy.

World Model	
Layers, width, heads, feed-forward	4, 128, 4, 256
Parameters	0.61 M
Unroll lengths U [steps]	48, 144, 288
Loss weights w_R, w_s, w_P	5, 0.3, 10
Memory time constants τ [steps]	12, 72, 288
Batch, steps per round	16 unrolls, 1000
Optimizer, learning rate	AdamW, $5 \cdot 10^{-5}$

Policy And DPC	
Layers, width, heads, feed-forward	3, 64, 4, 128
Parameters	0.12 M
Horizon H (curriculum) [steps]	144, 288
Discount γ, damping λ	0.995, 0.99
Comfort factor κ, filter α	1.8, 0.3
Batch B, iterations I	24 starts, 400
Optimizer, learning rate	AdamW, $3 \cdot 10^{-4}$

Algorithm 1 On-policy rounds of world-model training and DPC training of the policy.

$\mathcal{D} \leftarrow$ exploration episodes on the generated buildings; train f_ϕ on $\mathcal{D}$ with (6)
for round $n = 1, 2, \ldots$ **do**
  for iteration $i = 1, \ldots, I$ **do** ▷ DPC, warm-started from the previous policy
    $H \leftarrow$ curriculum horizon; sample an episode and B start indices t_0; $\hat{s}_{t_0} \leftarrow s_{t_0}$
    for $k = 0, \ldots, H-1$ **do**
      $\tilde{u} \leftarrow \alpha\, \pi_\theta(\hat{s}_{t_0+k}) + (1-\alpha)\, \tilde{u}$ (9); $\hat{s}_{t_0+k+1} \leftarrow f_\phi(\hat{s}_{t_0+k}, \tilde{u}, x_{t_0+k})$; memory (5)
      $\hat{r}_{t_0+k} \leftarrow$ reward rebuilt from $\hat{T}$ and $\hat{P}$ (7); damp the state gradient (8)
    end for
    $\theta \leftarrow \mathrm{AdamW}(\theta, -\nabla_\theta J)$ with clipped gradient norm, unless the loss is not finite
  end for
  $\mathcal{D}_n \leftarrow$ one 14-day episode per generated building with π_θ, partly with action noise
  fine-tune f_ϕ on $\mathcal{D} \cup \mathcal{D}_n$, half of every batch from $\mathcal{D}_n$; $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_n$
end for

4 Results

4.1 Evaluation Protocol

Policies are evaluated in the simulator on the 30 official training scenarios (six buildings under five weather stations), none of which is used for training, over 14 days (4032 steps), so that every episode includes a weekend and the following Monday start-up; earlier stages were evaluated over 7 days. The local 14-day score proved to be a reliable predictor of the Kaggle evaluation, with a ratio between the two of 1.23, 1.29 and 1.23 for the three policies submitted, against 1.39 and 1.48 for the 7-day protocol.

4.2 Overall Performance

The progression of the method is summarized in Figure 2, and its effect on the individual scenarios in Figure 3.

As can be observed in Figure 2, the first entity-token policy trained by DPC already reduces the score of the PID baseline from 0.0591 to 0.0155, that is, by 74%, and every following stage brings a further reduction, down to 0.0106 for the final policy, which corresponds to 0.013 on the Kaggle public leaderboard, against 0.019 for the first entity-token policy submitted. The largest single gain comes from the 24-hour horizon trained with the command filter in the loop (from 0.0133 to 0.0106).

As can be observed in Figure 3, the final policy is better in 23 of the 30 scenarios, with the largest reductions in the cold climates (Fairbanks 1 from 0.089 to 0.068, Fairbanks 8 from 0.046 to 0.027), while in the remaining scenarios the differences are within a few percent. Figure 4 shows how this is achieved in two heating scenarios with very different climates, namely the first week of January in Fairbanks 1, when the outdoor air drops below $-35\,^\circ$C, and the first week of November in Denver 8, with mild days and nights close to freezing.

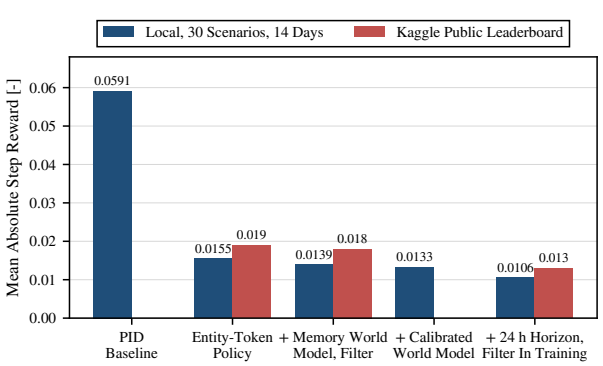


Figure 2: Mean absolute step reward of the shipped PID baseline and of the successive stages of the method, on the 30 official scenarios over 14 days and on the Kaggle public leaderboard.

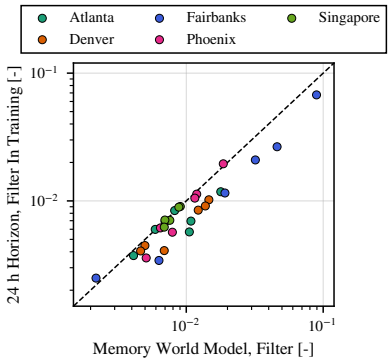


Figure 3: Score of each official scenario (14 days): filtered policy on WM-3 against the final policy; points below the diagonal are improvements.

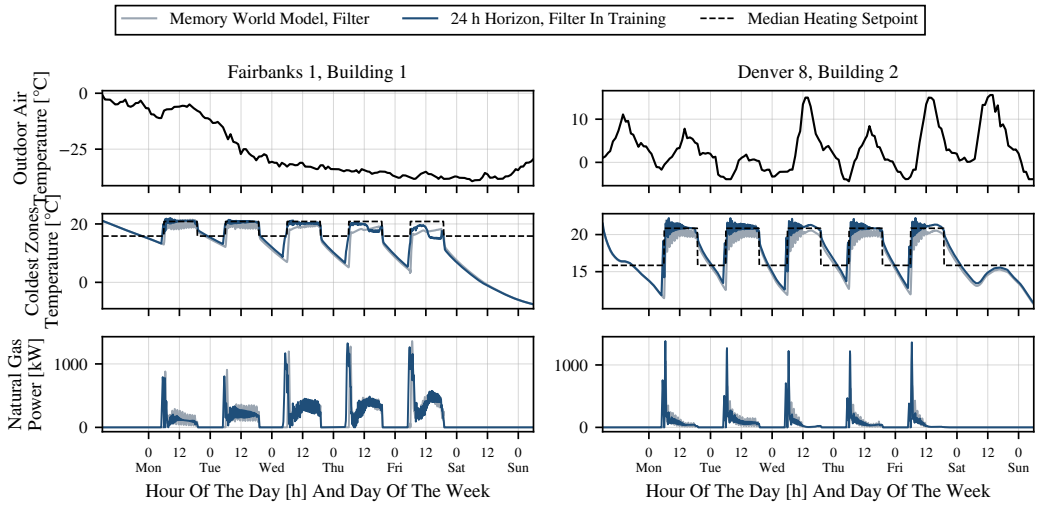


Figure 4: First week of Fairbanks 1 (official building 1, from 1 January) and of Denver 8 (official building 2, from 5 November): outdoor air temperature, 10th percentile of the zone temperatures and natural gas power of the filtered policy trained on WM-3 and of the final policy.

As can be observed in Figure 4, both policies switch the plant off at night and over the weekend, and the difference lies in the heat delivered during occupancy. In Fairbanks 1 the final policy heats more in the second half of each working day, thereby keeping the coldest zones closer to the setpoint, at the price of an 11% higher gas consumption, and its mean step reward over the week improves from -0.0650 to -0.0548. In Denver 8 the coldest zones of the filtered policy remain below the setpoint during the first hours of every working day, whereas those of the final policy reach it sooner, so that the comfort term of the week drops from 0.0121 to 0.0049 and the mean step reward improves from -0.0174 to -0.0121, despite a higher gas consumption. Over the 30 scenarios, the comfort term is more than halved (from 0.0065 to 0.0026), whereas the energy and carbon terms grow only slightly (from 0.0041 to 0.0043 and from 0.0033 to 0.0037, respectively).

4.3 World-Model Quality and Policy Quality

The world models were diagnosed on 45 clean episodes driven by a fixed policy, 30 on the official scenarios and 15 on unseen benchmark buildings, from 1080 start states, both by replaying the logged actions and in closed loop with the policy acting inside the model. Table 2 relates these diagnostics to the policies trained on each model.

Table 2: Diagnostics of the five world models (WM-1 to WM-5, Section 3.3) after 24 hours (temperature mean absolute error and bias, ratio of predicted to true comfort and energy terms) and 14-day score of the policies trained on each model.

World Model	Replay MAE [K]	Closed-Loop MAE [K]	Closed-Loop Bias [K]	Comfort Pred./True	Energy Pred./True	Policy (Horizon) 14-Day Score [-]
WM-1	1.38	1.13	$+0.43$	0.96	0.95	Entity-Token (6–12 h): 0.0155
WM-2	1.50	1.35	$+1.07$	0.92	1.09	–
WM-3	0.87	0.87	-0.18	1.22	1.22	Filter (6–12 h): 0.0139
WM-4	0.89	0.82	$+0.06$	1.00	1.10	Filter (6–12 h): 0.0133
						Filter In Training (24 h): **0.0106**
WM-5	0.78	0.79	-0.01	1.07	1.20	Filter In Training (24 h): 0.0121

As reported in Table 2, the long memory of WM-3 reduced the 24-hour error by about 35% with respect to WM-2, and the on-policy round of WM-4 removed both the temperature bias and the overestimation of discomfort (from 1.22 to 1.00). WM-5 has the lowest error of all, yet the same training recipe yields a worse policy on it than on WM-4 (0.0121 against 0.0106), better in only 7 of the 30 scenarios; WM-5 overestimates the comfort term by 7% and the energy term by 20%, and its policy spends more energy than needed, especially in the warm climates. This can primarily be attributed to the fact that the DPC gradient follows the predicted reward rather than the predicted temperature, so that a systematic bias on the reward terms steers the policy, whereas a random temperature error largely averages out.

4.4 Long Horizon And Command Filter

The step from 0.0133 to 0.0106 in Figure 2 combines the 24-hour horizon with the command filter applied inside the training rollouts, and the two elements are effective only together and only on an accurate world model. On WM-4, training with the filter but with the 6 to 12-hour horizon yields 0.0182, since the filtered commands respond more slowly and short rollouts do not reveal the benefit of acting earlier, whereas the long horizon lets the policy anticipate occupancy and cold nights while the filter keeps it from exploiting high-frequency errors of the model. On the older WM-2, whose closed-loop error after 24 hours is 1.35 K (Table 2), the same long horizon instead degraded the 7-day score from 0.0195 to 0.0328, since the policy exploited errors accumulated over the horizon.

5 Conclusion

This work has shown that a single policy for unseen buildings can be obtained without deep reinforcement learning and without imitation, by representing the building as a set of entities, learning a world model of its physics and differentiating the competition reward through closed-loop rollouts of that model. The resulting policy reduces the score of the shipped PID baseline by 82% on the official scenarios and reaches 0.013 on the Kaggle public leaderboard. The experiments indicate that the main driver of performance is the world model, in particular its calibration on the reward terms and its accuracy over the planning horizon. The next steps are the measurement of the variance between training seeds, ensembles of world models against the errors of any single model, and planning at deployment, in which the actions of the policy are refined online inside the world model.

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